

Exp 7

Use Hive to create, alter, and drop databases, tables, views, functions, and indexes.

Create a Database

```
CREATE DATABASE IF NOT EXISTS sample_db;
```

Create a Table

```
USE sample_db;
```

```
CREATE TABLE IF NOT EXISTS employee (  
    id INT,  
    name STRING,  
    salary FLOAT  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```

Alter a Table

```
ALTER TABLE employee ADD COLUMNS (dept STRING);
```

Drop a Table

```
DROP TABLE IF EXISTS employee;
```

Drop a Database

```
DROP DATABASE IF EXISTS sample_db CASCADE;
```

Create a View

```
CREATE VIEW IF NOT EXISTS high_salary_employees AS  
SELECT * FROM employee WHERE salary > 50000;
```

Create a Function (UDF)

Step 1: Write a Java Code

```
package com.custom.hive.udf;

import org.apache.hadoop.hive.ql.exec.UDF;
import org.apache.hadoop.io.Text;

public class ToUpper extends UDF {
    public Text evaluate(Text input) {
        if (input == null) return null;
        return new Text(input.toString().toUpperCase());
    }
}
```

Step 2: Create a jar file of above code.

Step 3: use UDF in Hive

-- Step 1: Add the JAR file to Hive session

```
ADD JAR /path/to/ToUpper.jar;
```

-- Step 2: Register the UDF using your custom class

```
CREATE TEMPORARY FUNCTION to_upper AS 'com.custom.hive.udf.ToUpper';
```

-- Step 3: Use the function in a HiveQL query

```
SELECT to_upper(name) FROM employee;
```

```
101,Rachel,72000.0,Finance
102,Marcus,51000.0,Engineering
103,Sophia,66000.0,HR
104,Daniel,48000.0,Support
105,Priya,59000.0,Marketing
106,Yusuf,81000.0,Engineering
107,Clara,43000.0,HR
108,Karan,55000.0,Finance
109,Linda,39000.0,Support
110,Wei,97000.0,Research
```

employee.csv

Create an Index

```
CREATE INDEX idx_salary ON TABLE employee (salary)
AS 'COMPACT'
WITH DEFERRED REBUILD;
```